Submodule 37.06: The Vanishing Gradient Problem (Hinglish)

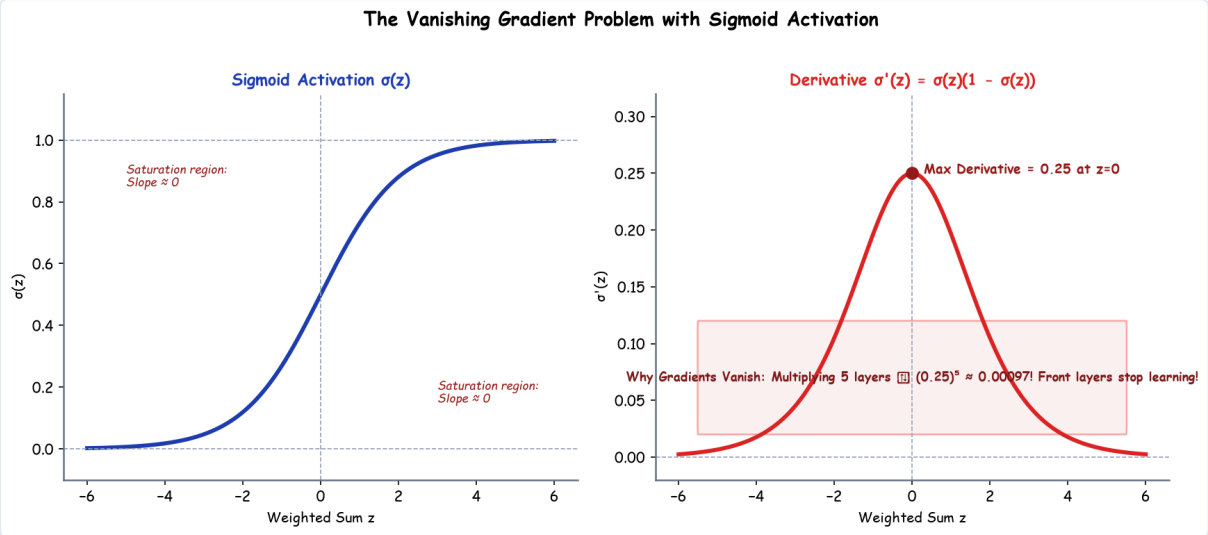
1. Vanishing Gradient Problem kya hai?

Jab hum deep neural network banate hain (jisme bohot saari hidden layers hoti hain), tab training ke dauran ek bohot badi problem aati hai jise **Vanishing Gradient Problem** kehte hain. Is problem me jaise-jaise error ka signal peeche ki layers (layer 1 aur 2) ki taraf jata hai, gradient ka size itna chhota ho jata hai ki wo lagbhag **0** ban jata hai!

Natija:

$$w_{new} = w_{old} - \eta \cdot (0.00001) \approx w_{old}$$

Pehli layers ke weights update hi nahi hote, aur network kuch seekh hi nahi pata!



---

2. Yeh Problem Kyun Aati Hai? (Root Cause)

Iska sabse bada kaaran hai purane activation functions (jaise **Sigmoid**):

1. **Sigmoid ka derivative bohot chhota hota hai**:

$$\sigma'(z) = \sigma(z)(1 - \sigma(z))$$

Iska **maximum derivative sirf 0.25** hota hai (jab $z = 0$ ho). Baki sab jagah yeh 0.25 se bhi chhota hota hai.

2. **Chain Rule me Continuous Multiplication**: Agar network me 5 hidden layers hain, toh gradient me 5 baar derivative multiply hoga:

$$(0.25)^5 = 0.00097$$

Itna chhota number multiply hone se gradient gayab (vanish) ho jata hai!

---

3. Iska Solution kya hai?

1. **ReLU Activation Use Karna**: Hidden layers me Sigmoid hata kar **ReLU** lagana (kyunki positive values ke liye ReLU ka derivative hamesha 1.0 rehta hai, kabhi vanish nahi hota).
2. **He / Kaiming Initialization**: Weights ko sahi variance ke saath shuru karna.
3. **Residual Connections (ResNet)**: Skip connections lagana taaki gradient seedhe bypass ho sake.\n